



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2024

Round 1: Online Knowledge Challenge

Senior Division (ages 15–18)

Mark scheme and worked solutions

Total: 60 marks. Accept answers within rounding and correct alternative methods. Apply error carried forward (ECF): if a wrong value from an earlier part is used correctly in a later part, award the method marks.

Part A: Answer key

Question	Answer	Correct option
A1	C	15.0 kWh
A2	C	1110 nm
A3	B	33%
A4	C	Rhizobium bacteria
A5	B	a large amount of biodegradable organic matter, for example from sewage
A6	C	818 MWh
A7	B	10 years
A8	A	About -18°C
A9	B	6.97 kWh
A10	B	Hydrogen made by electrolysis of water using renewable electricity

Part A: Worked solutions

A1. Answer C

Energy in $\times 0.90 = 13.5$ kWh, so energy in $= 13.5 / 0.90 = \mathbf{15.0}$ kWh. Option A wrongly multiplies by 0.90.

A2. Answer C

$\lambda = hc/E = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (1.12 \times 1.60 \times 10^{-19}) = 1.11 \times 10^{-6} \text{ m} = \mathbf{1110 \text{ nm}}$, in the near infrared. Longer wavelengths pass straight through.

A3. Answer B

Photons below the band gap are not absorbed, and the excess energy of high-energy photons is lost as heat. Together these limit a single junction to about **33%**. 86% is the limit for an infinite stack of junctions under concentrated light.

A4. Answer C

Rhizobium lives symbiotically in root nodules and converts N_2 into ammonium using the enzyme nitrogenase. Nitrifying bacteria oxidise ammonium to nitrate, and denitrifiers return nitrogen to the air.

A5. Answer B

BOD measures the oxygen that microorganisms consume while decomposing organic matter (usually over 5 days at 20°C). A high BOD means **a lot of organic pollution**, which will lower dissolved oxygen.

A6. Answer C

$m = 1.0 \times 10^9 \text{ kg}$. $E = mgh = 1.0 \times 10^9 \times 9.81 \times 300 = 2.94 \times 10^{12} \text{ J}$. Dividing by $3.6 \times 10^9 \text{ J}$ per MWh gives **818 MWh**.

A7. Answer B

Doubling time = $\ln 2 / \ln 1.07 = 10.2$ years. This matches the 'rule of 70': $70/7 = 10$ years.

A8. Answer A

Absorbed = emitted: $S(1 - \alpha)/4 = \sigma T^4$, so $T = (1361 \times 0.70 / (4 \times 5.67 \times 10^{-8}))^{1/4} = 255 \text{ K} \approx -18^\circ\text{C}$. The actual average of $+15^\circ\text{C}$ is 33 K warmer because of the natural greenhouse effect.

A9. Answer B

$Q = mc\Delta T = 150 \times 4.18 \times 40 = 25\,080 \text{ kJ}$. Dividing by 3600 kJ per kWh gives **6.97 kWh**. Option C is the answer in MJ, not kWh.

A10. Answer B

Green hydrogen comes from electrolysis powered by renewables. A is 'blue', C is 'brown/black' and D is 'grey' hydrogen, which is how most hydrogen is made today.

Part B: Worked solutions and mark scheme

B1. Reusable or single-use? (9 marks)

(a) [3 marks] Need $(0.40 + 0.03n)/n < 0.17$, so $0.40 < 0.14n$ and $n > 2.86$. The bottle must be used at least **3 times**.

Marking: 1 mark for the inequality; 1 mark for solving it; 1 mark for rounding up to a whole number.

(b) [2 marks] $(0.40 + 20 \times 0.03)/20 = 1.00/20 = \mathbf{0.050 \text{ kg}}$ per use. Saving = $(0.17 - 0.050)/0.17 = \mathbf{71\%}$.

Marking: 1 mark each.

(c) [4 marks] A bottle is used exactly k times with probability $p^{k-1}(1-p)$. So $E[k] = \sum k p^{k-1}(1-p) = (1-p)/(1-p)^2 = 1/(1-p) = \mathbf{10 \text{ uses}}$. Footprint per use = $(0.40 + 10 \times 0.03)/10 = \mathbf{0.070 \text{ kg CO}_2\text{e}}$, still well below the can.

Marking: 1 mark for the geometric distribution; 1 mark for summing the series (or an equivalent argument); 1 mark for 10 uses; 1 mark for 0.070 kg.

B2. A solar farm (8 marks)

(a) [2 marks] $50 \text{ MW} \times 8760 \text{ h} \times 0.18 = 78\,840 \text{ MWh} = \mathbf{78.8 \text{ GWh}}$.

Marking: 1 mark for the method; 1 mark for the answer.

(b) [2 marks] $78\,840\,000 \text{ kWh} / 3500 \text{ kWh} = \mathbf{\text{about } 22\,500 \text{ households}}$.

Marking: Allow error carried forward.

(c) [2 marks] $78\,840\,000 \times 0.95 = 7.49 \times 10^7 \text{ kg} \approx \mathbf{74\,900 \text{ t}}$ of CO_2 .

Marking: Allow error carried forward.

(d) [2 marks] Any two of: no output at night; clouds; low sun angle in winter and in the early morning and evening; panels lose efficiency when hot; dust and soiling; inverter and cable losses.

Marking: 1 mark each.

B3. Sewage effluent in a river (8 marks)

(a) [2 marks] $\text{BOD} = (4.0 \times 2.0 + 0.50 \times 60)/(4.0 + 0.50) = 38/4.5 = \mathbf{8.4 \text{ mg L}^{-1}}$.

Marking: 1 mark for the flow-weighted mean; 1 mark for the answer.

(b) [2 marks] $\text{DO} = (4.0 \times 9.0 + 0.50 \times 2.0)/4.5 = 37/4.5 = \mathbf{8.2 \text{ mg L}^{-1}}$.

Marking: 1 mark for the method; 1 mark for the answer.

(c) [2 marks] The oxygen demand (8.4) exceeds the oxygen available (8.2), so DO would fall to **zero**: the river would become anoxic and fish would die.

Marking: 1 mark for comparing the values; 1 mark for the conclusion.

(d) [2 marks] Allowed mixed BOD = $37/4.5 - 5.0 = 14.5/4.5 \text{ mg L}^{-1}$. So $(8.0 + 0.50x)/4.5 \leq 14.5/4.5$, giving $0.50x \leq 6.5$ and $x \leq \mathbf{13 \text{ mg L}^{-1}}$.

Marking: 1 mark for the inequality; 1 mark for the answer.

B4. Fertilisers and nitrous oxide (8 marks)

(a) [2 marks] NH_4NO_3 : $28.02/80.05 = \mathbf{35.0\%}$. Urea: $28.02/60.06 = \mathbf{46.7\%}$.

Marking: 1 mark each.

(b) [2 marks] N needed = $120 \times 15 = 1800$ kg. Urea = $1800/0.4665 = 3858$ kg $\approx$ **3.86 t**.

Marking: 1 mark for total N ; 1 mark for the answer.

(c) [3 marks] $N_2O-N = 18$ kg. Mass of $N_2O = 18 \times 44.02/28.02 = 28.3$ kg. $CO_2e = 28.3 \times 273 = 7720$ kg $\approx$ **7.7 t CO_2e** .

Marking: 1 mark for 18 kg; 1 mark for converting N to N_2O ; 1 mark for CO_2e .

(d) [1 mark] For example: split applications that match crop uptake; precision (variable-rate) spreading; nitrification inhibitors; legumes in the rotation to fix nitrogen naturally.

Marking: 1 mark.

B5. How long will it last? (7 marks)

(a) [1 mark] $880/26 =$ **34 years** (33.8).

Marking: 1 mark.

(b) [4 marks] $880 = (26/0.030)(e^{0.030t} - 1)$, so $e^{0.030t} = 1 + 0.030 \times 880/26 = 2.015$. Then $t = \ln(2.015)/0.030 =$ **23 years**. Steady 3% growth cuts the lifetime by a third.

Marking: 1 mark for setting up the equation; 1 mark for rearranging; 1 mark for the logarithm; 1 mark for the answer.

(c) [2 marks] Any two of: higher prices make lower-grade deposits economic, so reserves grow; new discoveries; recycling; substitution by other materials; more efficient use.

Marking: 1 mark each.